

Radiation Effects from Energetic Heavy Ions: Damage from Particles Accelerated by Interplanetary Shocks via Simulations

Ions are accelerated to high energies ($> \text{MeV}$) by shocks associated with coronal mass ejections. Single Event Effects from these high mass Solar Energetic Particles present unique challenges to spacecraft electronics. However, energetic ions are relatively diffuse and do not strongly influence the bulk flow of the plasma. As such, it is inefficient temporally and computationally to model the behavior of SEPs via conventional methods such as Particle-In-Cell or Hybrid codes – new, cutting-edge methods are needed. We propose a new simulation suite to predict heavy ion SEP fluxes using kinetic models coupled to a deterministic radiation transport code. The proposed simulations will generate differential intensities for heavy ions up to $Z = 26$ and use the data for rapid forecasts of Single Event Effect failure rates. Our suite will be constrained four observation-based parameters: Solar Wind speed, shock speed, plasma beta, and Solar Wind magnetic field strength. A minimized set of inputs will allow for an optimized, yet rigorous simulation suite. The proposed program will provide critical data streams for operational use of space assets exposed to the natural space radiation environment. Completion of this project will result in simultaneous reduction design risk and cost for future deployments.

- **Scientific Justification**

Section 1: Ionizing Radiation from IP Shocks damage space-based electronics

The interaction between high energy charged particles, photons, and neutrons and active electronic devices is known to induce malfunctions in or cause catastrophic damage to the system (Peterson, 2011). In the near-Earth space and the high atmosphere environment space-based assets are particularly exposed to i) cosmic rays, ii) trapped radiation in the geomagnetic field, and iii) energetic particles from solar storms and eruptions. The cosmic ray and trapped radiation environments, while variable with Solar Activity, can be predicted with high accuracy (Selesnick & Albert, 2019; Slaba & Whitman, 2020). However, the impact of coronal mass ejections and solar flares from the Sun are more difficult to forecast.

Interplanetary shocks associated with CMEs accelerate high intensities of protons, electrons and ions into the Earth's magnetosphere. Solar Energetic Particles (SEPs) represent the major radiation hazard to spacecraft (Liu et al, 2026). Current forecasting methods rely on predicting peak fluxes within a given confidence interval. Other, more robust, methods ignore heavy ions entirely due to computational constraints (Liu et al, 2026). Heavy ions are ignored due to the difficulty in modeling the behavior of trace particles in conventional self-consistent plasma simulation paradigms. Accurate modeling is critical for assessing risk in the design of future mission systems.

The effect of ionizing radiation on matter is known to depend on the kinetic energy of the incident charged nucleon (Carron, 2021). The linear energy transferred, naturally, scales exactly with the mass number and charge state of the particle. The failure rate of a device due to radiation, equation 1, shows this behavior. A radiation transport kernel $K(E)$ is linked to the environment, the differential flux $j(E, \vec{x}, t)$, by the incident energy E .

$$R(\vec{x}, t) = \int K(E)j(E, \vec{x}, t)dE$$

Current models either ignore important event to event variation for SEP ion fluxes or exclude the behavior. We propose a new simulation suite which will couple efficient models of heavy ion acceleration by interplanetary shocks with deterministic radiation transport code. The new simulation suite will provide accurate SEP fluxes up to $Z=26$ from observationally source shock parameters: Θ_{Bn} , U_{shock} , $\frac{B_2}{B_1}$, & β . These outputs will flow directly into the radiation transport code and calculate Single Event Effect failure rates in real-time.

Section 2: Modeling Heavy Ion acceleration computational challenging

Ions are accelerated to high energies ($> \text{MeV}$) by shocks associated with coronal mass ejections, producing Solar Energetic Particles that can cause Single Event Effects in spacecraft electronics. Since heavy ions only make up a small portion of the total mass and pressure of the CME (Desai & Giacalone, 2016), it is difficult to include them into conventional, self-consistent simulation techniques. Heavy Ion fluxes are partitioned based on solar composition (Robinson et al, 2020). This assumption leaves much to be desired. It is known that the ion flux partition varies from event to event (Desai & Giacalone, 2016). Accurate prediction SEP fluxes and associated radiation effects will

improve the reliability of spacecraft, reduce cost for future missions. This effort requires a robust simulation technique that can resolve the physics of the heavy ion acceleration.

We propose a simple test particle simulation for modeling the behavior of heavy ions at shocks. Test particles and kinematic simulations have already been shown to accurately represent the physics of diffusive shock acceleration of ions (Reese & Giacalone, 2026). The radiation effects simulation will primarily predict single event effect failure rates. The kinematic simulation will provide energy spectra and fluxes of heavy ions SEPs which can be used as inputs for the coupled radiation effects operations analysis tool.

The near-Earth space environment can vary greatly depending on system trajectory. There is no one size fits all radiation hardness design. Over-designing can be expensive – a difficult constraint for many scientific endeavors. Accurate forecasting of the specific radiation exposure to a system on its’ own flight path can greatly reduce cost while also maintaining appropriate risk posture. This proposal represents a first time coupling of environments with deterministic radiation transport to efficiently and rigorously predict heavy-ion SEP fluxes and SEEs. The coupled simulation proposal will allow for the development of robust shielding early in the design phase, assess appropriate concept of operations, and (if necessary) swap-out highly susceptible parts.

Section 3: This Proposal: Coupled Simulation Framework for SEP Forecasting

The customer community has identified the need for state-of-the-art natural space radiation environment and single event effects modeling. Our proposal addresses the need for operational analysis tools for satellites and space vehicles. This tool will provide spacecraft anomaly assessment capabilities for mission operators. The assessment capabilities will leverage SEP flux simulations for use in analysis workflow.

We propose a kinetic modeling framework for heavy ion fluxes that is explicitly grounded in observable heliophysical parameters, including shock speed, plasma beta, and upstream solar wind conditions such as magnetic field strength and density. By anchoring the model to these measurable quantities, the approach ensures both physical fidelity and direct applicability to real-time or forecasted space weather inputs. The kinetic treatment captures the energy-dependent transport and acceleration of heavy ions, enabling accurate prediction of differential fluxes across species and charge states. Crucially, these flux outputs are directly coupled to a deterministic failure rate calculation, allowing for rapid assessment of Single Event Effect risks in spacecraft systems. This integrated framework bridges first-principles plasma physics with engineering-relevant reliability metrics, providing a practical and predictive tool for space environment hazard assessment.

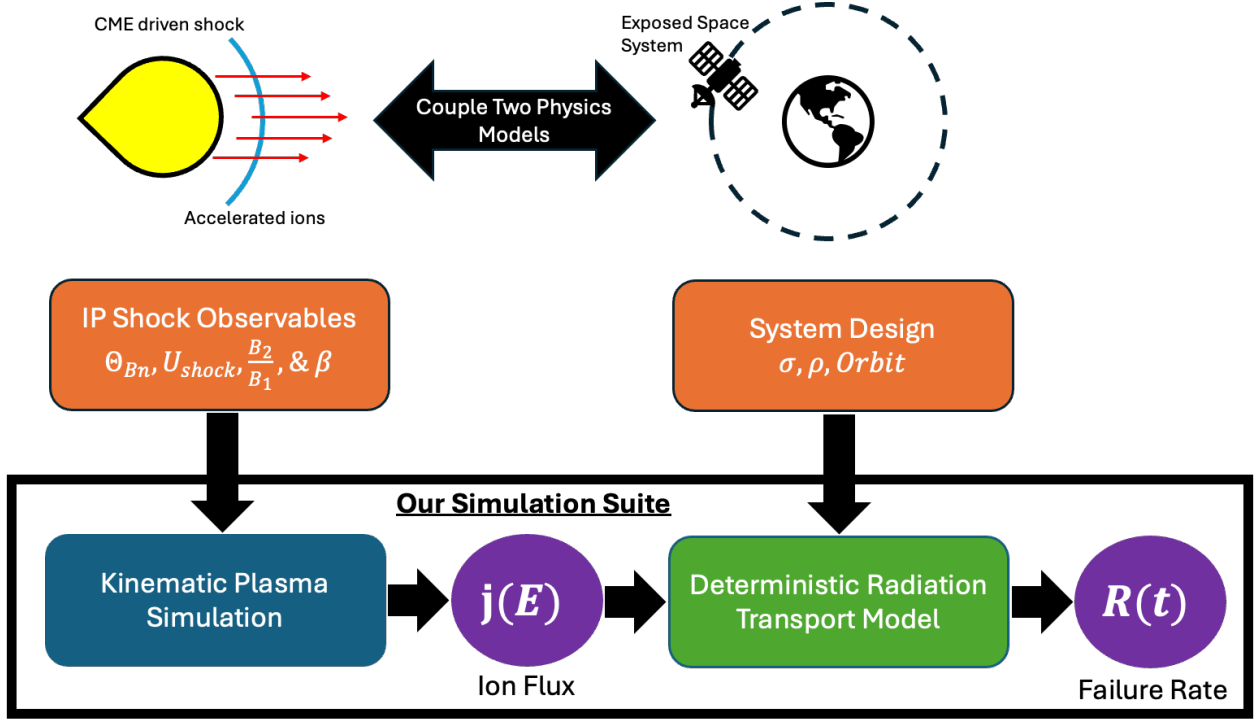


Figure 1: **Flow chart of the proposed simulation method.** The top sketches the two existing paradigms of modeling the physics of ion acceleration and the physics of radiation effects separately. The observables related to both systems can instead be leveraged by our model. Our proposal will chain a kinematic simulation for ion acceleration to a deterministic radiation transport code to predict radiation hardness of future mission designs.

Section 4: Simulation and Prediction reduce risk posture in designs

The natural space radiation environment is global. To protect assets on Earth and in space, we need to be able to predict the radiation environment from the Sun with great accuracy. Variability in the space environment impacts all technology in space. The proposed program will provide critical data streams for operational use of space assets exposed to the natural space radiation environment. Completion of this project will result in simultaneous reduction design risk and cost for future deployments. The heliophysics decadal (NASEM 2024) states that space weather research needs to evolve from interpretive science into a predictive discipline. Space weather research is driven by observations, validated physics-based models, and direct coupling to user applications. Our proposal is operationally integrated and addresses these key requirements. It provides actionable quantitative risk assessment by translating fundamental plasma physics into metrics useful to actual engineers. Our coupled simulation suite enables a direct research-to-operations (R2O) pipeline which addresses the exact missing capabilities highlighted in the decadal survey.

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